

Predictive Sorting of Cryptocurrencies Based on Fundamentals and Sentiment*

Massimo Guidolin[†], Serena Ionta[‡]

December 30, 2024

Abstract

We study the relationship between cryptocurrency returns and fundamental blockchain characteristics, with a special emphasis on the predictive power of fundamentals and sentiment. In particular, we perform a systematic investigation of the predictive power of blockchain characteristics, such as the variation in the hashrate and the rate of growth of active users. We evaluate the effectiveness of these variables in forecasting returns for a large sample of cryptocurrencies, using metrics such as the out-of-sample R^2 and certainty equivalents to validate the model's reliability. Additionally, we use standard sorting methodologies in empirical asset pricing that identify portfolios that invest in cryptocurrencies with higher predictability while shorting those with lower predictability. This sorting strategy yields promising portfolio spreads. This emphasizes the predictive value potential of blockchain fundamentals and sentiment indicators in cryptocurrency markets. Our findings suggest that a combined approach based on both technical and behavioral factors provides robust insights into cryptocurrency price dynamics.

Keywords: Cryptocurrencies, Blockchain Fundamentals, Investment Sentiment, Hashrate, Out-of-Sample, Forecasting Risk Factors, Portfolio Sorting.

JEL Codes: G11, G12, G17, E44, D53.

*We thank the participants of various conferences and seminars for their suggestions. We are also grateful to Antonio Afonso, Daniele Bianchi, Giovanna Ciaffi, Maximilian Goebel, Francesco Giuli, Antonino Lofaro, Paolo Matteucci, Alex Orlov, Manuela Pedio and Lorenzo Tonni for their valuable comments. Carlo Benanati provided excellent research assistant in deriving preliminary results. This study was funded by the European Union - NextGenerationEU, in the framework of the GRINS - Growing Resilient, INclusive and Sustainable project (GRINS PE000018 - CUP B43C22000760006). The views and opinions expressed are solely those of the authors and do not necessarily reflect those of the European Union, nor can the European Union be held responsible for them.

[†]Corresponding Author. Department of Finance. Bocconi University. Email: massimo.guidolin@unibocconi.it

[‡]Department of Finance. Bocconi University. Email: serena.ionta2@unibocconi.it

Predictive Sorting of Cryptocurrencies Based on Fundamentals and Sentiment*

December 30, 2024

Abstract

We study the relationship between cryptocurrency returns and fundamental blockchain characteristics, with a special emphasis on the predictive power of fundamentals and sentiment. In particular, we perform a systematic investigation of the predictive power of blockchain characteristics, such as the variation in the hashrate and the rate of growth of active users. We evaluate the effectiveness of these variables in forecasting returns for a large sample of cryptocurrencies, using metrics such as the out-of-sample R^2 and certainty equivalents to validate the model's reliability. Additionally, we use standard sorting methodologies in empirical asset pricing that identify portfolios that invest in cryptocurrencies with higher predictability while shorting those with lower predictability. This sorting strategy yields promising portfolio spreads. This emphasizes the predictive value potential of blockchain fundamentals and sentiment indicators in cryptocurrency markets. Our findings suggest that a combined approach based on both technical and behavioral factors provides robust insights into cryptocurrency price dynamics.

Keywords: Cryptocurrencies, Blockchain Fundamentals, Investment Sentiment, Hashrate, Out-of-Sample, Forecasting Risk Factors, Portfolio Sorting.

JEL Codes: G11, G12, G17, E44, D53.

1 Introduction

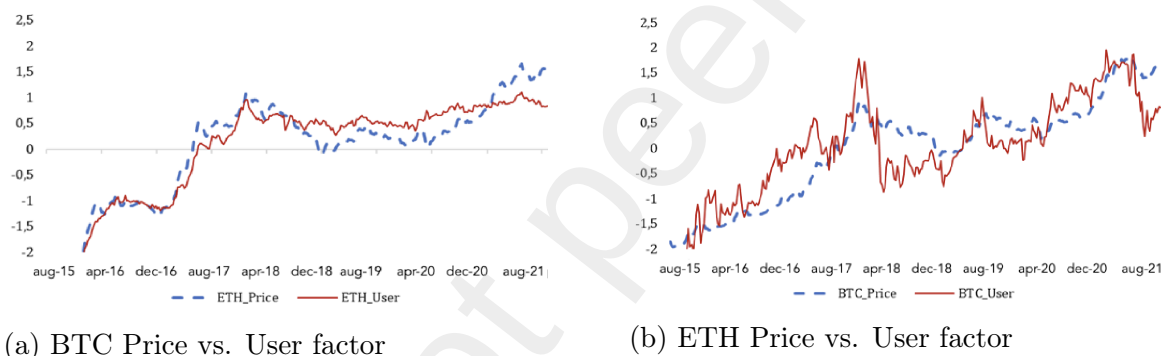
The exponential growth and extreme swings in the cryptocurrency markets have progressively made the need to understand the fundamental drivers of cryptocurrency valuations more acute. When compared to traditional asset classes, cryptocurrencies are characterized by fewer identifiable fundamental metrics; this makes their valuation particularly challenging. While traditional finance has long relied on cash dividends, earnings, free cash flows and other company-specific fundamental notions to price traditional assets (such as stocks and bonds, [Gordon \(1959\)](#); [Campbell and Shiller \(1988\)](#)), the valuation of cryptocurrencies often hinges on blockchain-specific fundamentals such as network activity and security. Additionally, recent insights from behavioral finance have emphasized the importance of sentiment-based variables in explaining asset price deviations from their underlying, rational values ([Shiller, 1981](#); [Baker and Wurgler, 2006](#); [Stambaugh et al., 2012](#)). Although a number of theoretical advancements concerning the relationship between blockchain fundamentals and cryptocurrencies have occurred, the systematic, empirical validation of their importance has remained limited. As a result, our study aims to bridge this gap by offering robust, data-driven evidence to support cryptocurrency pricing models that leverage on the features of the underlying blockchain. We systematically investigate the role played by both fundamental and sentiment-based factors in the modelling of cryptocurrency monthly returns. Specifically, we consider the blockchain hashrate and the active user count as fundamental characteristics; we also resort to Google Trends data to capture market sentiment ([Pagnotta and Buraschi, 2018](#); [Biais et al., 2023](#)). A cryptocurrency's hashrate, expressed in terahashes, serves as a proxy for the resources miners commit to securing the network; the active user count reflects instead the size of the cryptocurrency network. Sentiment, measured through Google search trends, complements these fundamentals by capturing fluctuations in market interest for the sphere of digital currencies ([Liu et al., 2022](#)).

Our data cover the period from August 2015 through September and initially cover a sample of five major cryptocurrencies: Bitcoin (BTC), Ethereum (ETH), Litecoin (LTC), Monero (XMR), and Dash (DASH). These cryptocurrencies together represent a substantial portion of the total market capitalization, accounting for approximately 70% to 90% the total market capitalization of the cryptocurrency market during the analysis period. Initially, the analysis focuses on these high-market-cap cryptocurrencies, which rely on Proof of Work (PoW) consensus mechanisms. However, in a subsequent stage of the anal-

ysis, the sample is expanded to include a total of 41 cryptocurrencies, to perform sorting analysis and out-of-sample (henceforth, OOS) tests.¹

Figures 1 - 3 display the weekly logarithmic prices of BTC and ETH, arguably the two oldest and most traded cryptocurrencies worldwide, along with the corresponding User, HashRate, and Google factor indices, respectively, to be constructed later. Visibly, we observe a positive relationship between prices and the three fundamental factors examined in this paper. Interestingly, in the case of Figures 1 and 3, in the case of, the user and sentiment factors even appear to move at a higher frequency than the price index does. Apart from providing a sense for the dynamics of the phenomena under investigation over time, these plots immediately reveal rather stark and precise connections between cryptocurrency prices and blockchain characteristics (as well as sentiment).

Figure 1
Cryptocurrency prices vs. the User factor

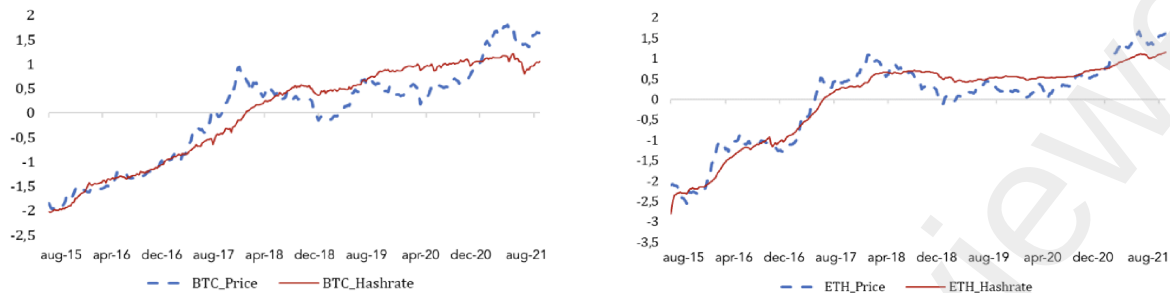


The charts represent the weekly logarithm of the price of BTC and ETH and the associated log-User index factor. The two panels report standardized values (the original value minus the mean, everything divided by the standard deviation). The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations.

Our paper builds on the work by [Bhambhwani et al. \(2023\)](#) reviewed in Section 2, but we contribute to the literature in at least three ways. First, we document the predictive power of blockchain fundamentals for cryptocurrency returns, even when we control for or compare to with the predictive power of momentum or sentiment. Second, we introduce a portfolio sorting methodology that follows the tradition in asset pricing studies. This methodology is specifically designed to address common challenges in the determination of

¹Despite this expanded sample, the lack of reliable hashrate-type data for newer blockchain types not based on standard POW protocols, limits to 41 the number of cryptocurrencies. In particular, all those digital currencies that use alternative consensus mechanisms such as Proof of Stake (PoS) or Delegated Proof of Stake (DPoS) are excluded.

Figure 2
Cryptocurrency prices vs. the HashRate factor

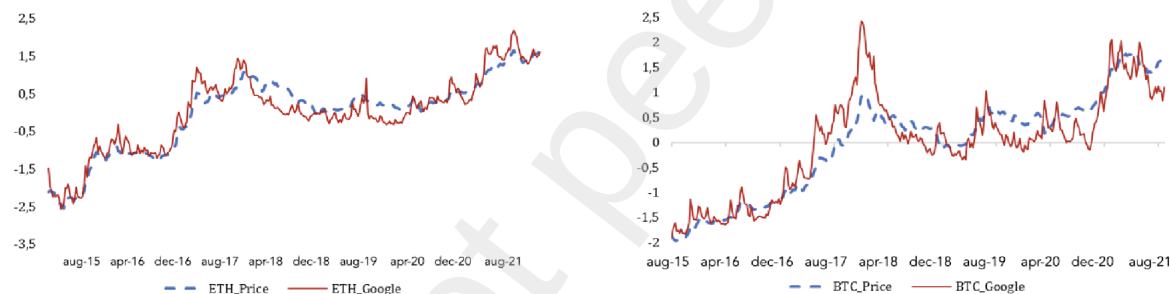


(a) ETH Price vs. the HashRate factor

(b) BTC Price vs. the HashRate factor

The charts represent the weekly logarithm of the price of BTC and ETH and the associated log-HashRate index factor. The two panels report standardized values (the original value minus the mean, everything divided by the standard deviation). The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations.

Figure 3
Cryptocurrency prices vs. the Google Sentiment factor



(a) ETH Price vs. the Google Sentiment factor

(b) BTC Price vs. the Google Sentiment factor

The charts represent the weekly logarithm of the price of BTC and ETH and the associated log-Google search index. The two panels report standardized values (the original value minus the mean, everything divided by the standard deviation). The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations.

factors driving prices, such as non-linearities, misspecification biases, and multicollinearity, which are instead frequently plaguing traditional recursive regression models (Bali et al., 2016; Cattaneo et al., 2020; Kagkadis et al., 2023). Third, we extend Bhambhwani et al. (2023)'s work by incorporating investment sentiment as a core component of the analysis, both in- and out-of-sample.

By integrating blockchain fundamentals with sentiment data, our study evaluates their predictive power using both classical statistical metrics for predictive accuracy and certainty

equivalent return estimates derived from classical mean-variance set ups. The proposed sorting framework effectively identifies assets with high vs. low predictability, and by systematically investing in (selling short) cryptocurrencies that are highly (scarcely) predictable, it consistently delivers positive abnormal returns. Within our recursive OOS design, these findings emphasize the importance of the fundamental factors in forecasting cryptocurrency prices.

The first part of the analysis employs two-factor and three-factor linear regression models to assess the influence of Bitcoin and momentum factors on cryptocurrency returns. While Bitcoin's impact weakens when blockchain-specific variables are included, it remains significant in simpler models. Adding blockchain-related characteristics improves explanatory power, enhancing the models' overall performance. Yet, the main contribution of our paper lies in the second part, in which we systematically assess the predictive capabilities of blockchain factors and sentiment data in a OOS context. Long-short portfolios constructed based on predictive performance show a notable increase in alpha when the long-short portfolios are built on the basis of predictability assessed through certainty equivalent returns, as opposed to OOS R^2 . The portfolio sorting method, consistently achieves positive and large returns, supporting the existence of remarkable economic value in combining blockchain fundamentals with sentiment indicators to understand cryptocurrency market dynamics.

Apart from creating a connection between predictability and a viable investment strategy, our paper therefore emphasizes the superior value of risk-adjusted measures in assessing predictive accuracy. Moreover, our methodology resorts to typical techniques in the asset pricing literature that naturally connect the size of the risk premia derived from predictability to the variability of the stochastic discount factor (SDF) induced by the predictors and especially the correlation of the SDF with the predictors (see, e.g., [Kan and Zhou \(2007\)](#)). Hence, assuming the absence of segmentation, our empirical findings of substantial predictability from fundamental blockchain variables to asset returns and of the associated cross-sectional risk premia, are consistent with the fact that such technological parameters may have become pervasive enough to affect the vary nature of the fundamental pricing measure characterizing the financial markets (under completeness).

The rest of the paper is organized as follows. Section 2 reviews relevant literature on cryptocurrency valuation and its predictability. Section 3 provides details on the data and on the methodology. Section 4 presents the empirical results, and Section 5 concludes.

2 Literature Review

Because the literature on cryptocurrencies is too vast to be accommodated in a few pages (see, instead, the excellent surveys by [Zheng et al. \(2018\)](#), [Härdle et al. \(2020\)](#), [Valdeolmillos et al. \(2020\)](#), and [Allen et al. \(2022\)](#)), we devote this section to a review of the main findings concerning the role played by the fundamental characteristics of blockchain technology in the asset pricing of digital currencies and, more generally, on positioning cryptocurrencies in relation to traditional asset classes, especially standard currencies, see, e.g., [Mohamad and Stavroyiannis \(2022\)](#) and [Xu and Kinkyo \(2023\)](#).

For instance, [Yermack \(2024\)](#) has recently argued that Bitcoin lacks comparability with traditional currencies, given its limited use as a medium of exchange and its volatility undermining its reliability as a store of value. However, this view may now warrant partial reconsideration, as a growing number of businesses and financial institutions are adopting cryptocurrency payment systems.² Another argument used to distinguish cryptocurrencies from conventional currencies is the lack of a secondary market for the efficient and secure trading of crypto assets. Research has particularly focused on the peculiar features and risks of the exchanges where these cryptocurrencies have been, and continue to be, traded. For instance, [Makarov and Schoar \(2020\)](#) analyzed the prices of Bitcoin, Ethereum, and Ripple, finding discrepancies across different exchanges for the same cryptocurrencies, leading to potential arbitrage opportunities and pointing to a lack of either liquidity or of adequate technological platforms able to guarantee fair pricing.³

In a different vein, [Baur et al. \(2018\)](#) examined the statistical properties of Bitcoin returns and found that it exhibited no correlation with traditional asset classes, such as stocks, bonds, and commodities. This occurred during both periods of stability and in times of financial crises. Their analysis concluded that Bitcoin was still primarily used as a speculative asset rather than a viable alternative to conventional currencies. In the context of further asset-driven comparisons, [Selgin \(2015\)](#) had in fact already suggested that Bitcoin shared certain characteristics with commodities, particularly because of its

²For instance, in 2021 Tesla announced its willingness to accept Dogecoin as a means of payment of some purchases. For details, see [Bloomberg, 2021](#).

³These price inefficiencies were found to often persist for extended periods, sometimes lasting for several weeks. Furthermore, the security of these platforms has historically relied on algorithms created and managed by third parties, without authoritative oversight to ensure safety and compliance, raising concerns over potential frauds. A well-known case is the Canadian exchange QuadrigaCX, which, at its peak, was considered to be the largest cryptocurrency market in Canada but then turned out to be a scam valued at approximately 190 million USD, see [Telford \(2019\)](#) and [Yousaf et al. \(2023\)](#).

programmed scarcity.⁴ He observed that while the supply of Bitcoin will continue to increase until 2040, it will eventually resemble a "synthetic commodity", with a supply that remains stable or gradually decreases thereafter.

Foley et al. (2019) estimated that between 2009 and 2017, approximately 46% of Bitcoin transactions were linked to illicit activities, though this fraction declined significantly after 2013. The historical evolution of Bitcoin suggests that its value and usage have diverged from criminal activities, as its price continued to rise even after the decline in illicit transactions, reinforcing the view that such activities are not a primary value driver for digital currencies.

An emerging literature has sought to identify market factors that can explain cryptocurrency returns, once more in comparison with other asset classes. Liu and Tsyvinski (2021) identified two key drivers: a momentum factor based on historical cryptocurrency returns and an investor attention factor that captures popular interest in cryptocurrencies using data from platforms like Google and Twitter. Liu et al. (2022) reached similar conclusions, expanding their research to include a broader range of cryptocurrencies. In related research, Bianchi et al. (2023) have investigated the predictability of returns in cryptocurrency markets and the forecasting power of market risk factors, cryptocurrency-specific predictors, and sentiment variables for digital currencies compared to traditional asset classes. Using a dynamic model to account for time-varying coefficients and model selection, they find that unlike traditional assets, cryptocurrencies are not systematically predicted by stock market factors or commodity prices, but exhibit significant exposure to investor attention.⁵ Bianchi et al. (2022) focused on trading dynamics and found that price variations and trading volumes offer valuable information to forecast price movements, in a manner that is rather unique to cryptocurrencies. The pioneering paper by Easley et al. (2019) focused on Bitcoin's blockchain mechanics. Their work explored the interactions between the mining reward algorithm, the transaction fees, and the waiting times associated with recording transactions on the blockchain ledger. They found that as Bitcoin prices rise, transaction fees increase, which in turn limits blockchain capacity and

⁴Programmed scarcity refers to the design choice made in many blockchain-based systems to limit the total supply of a cryptocurrency or control its issuance rate. Scarcity is typically encoded into the cryptocurrency's protocol and enforced by its decentralized network; it is a feature intended to promote long-term value stability. The controlled issuance rate instead ensures a predictable supply curve. For instance, Bitcoin carries a fixed upper limit on the number of coins (21 million) that can be created.

⁵This lack of predictability compared to traditional assets, combined with the benefits of diversification, results in realised expected utility gains for a power utility investor when cryptocurrencies are used in portfolio diversification.

user participation, hindering Bitcoin’s development as a transaction medium—a trend that has been observed in other cryptocurrencies, such as Ethereum and Dogecoin.

Finally, as already mentioned in the introduction, a study closely related to ours is [Bhambhwani et al. \(2023\)](#), who investigated whether blockchain characteristics, such as network size (measured by the number of unique active addresses) and computing power (measured by hashrate, i.e., resources used in proof-of-work confirmations), influence the returns of 18 major cryptocurrencies over 2017–2021. Using both cointegration tests and distributed lag regressions, they found significant positive long-term trends between prices, network size, and computing power for most PoW cryptocurrencies. At the aggregate market level, these factors explained the variation in expected cryptocurrency returns, performing as well as, if not better than, traditional return-based models. This evidence reinforces the view of certain cryptocurrencies as digital commodities, with prices shaped by production and usage factors.⁶

Our paper takes steps from [Bhambhwani et al. \(2023\)](#) but contributes to the literature by adopting an OOS perspective and by focusing on a portfolio sorting methodology that is robust to the existence of non-linearities, to any generic functional misspecification, and to multi-collinearity, that plague instead classical, recursive regressions in asset pricing research (see [Cattaneo et al. \(2020\)](#), [Kagkadis et al. \(2023\)](#), or [Bali et al. \(2016\)](#) for a textbook treatment). Specifically, we assess how the fundamental factors predict cryptocurrency *returns* using metrics like R^2 and the CER derived from managed portfolios resulting from systematic, recursive sortings. A novel portfolio sorting characteristic-based method is employed, which invests in assets with high predictability, consistently yielding positive performance. This emphasizes the relevance of blockchain fundamentals and sentiment in forecasting cryptocurrency returns.

2.1 Fundamental Characteristics of Blockchains

Blockchain, originally conceived for the validation of digital documents, is a technology that facilitates the creation of decentralized and immutable ledgers.⁷ It consists of a "block" containing data, a unique hash, and the hash of the previous block. These are linked in a chain. While a blockchain by itself is not a cryptocurrency, it serves as

⁶In [Bhambhwani et al. \(2023\)](#), the blockchain-based factors explained the variation in expected cryptocurrency returns at least as well as, if not better than, the return-based models.

⁷They are digital record-keeping systems that store transaction data in a secure and transparent way.

the underlying technology for many cryptocurrencies, enabling decentralized transaction validation.

Cryptocurrencies are either mineable, like Bitcoin, and hence rely on miners solving puzzles, or non-mineable tokens, pre-allocated through protocols (Cong et al., 2021). The key distinction between the two types lies in the consensus mechanism, which governs transaction validation and blockchain operations. The most widely used algorithms are Proof of Work (PoW) and Proof of Stake (PoS). PoW, the oldest and most recognized, relies on miners' computational work to validate transactions and secure the network. Although slow and energy-intensive, it is considered the most decentralized and secure mechanism (Chaudhry and Yousaf, 2018). PoS, by contrast, replaces miners with validators who stake their assets to validate transactions, improving speed and energy efficiency while reducing decentralization costs (Cong et al., 2021).⁸

The computational power, or hashrate, is a crucial metric to size up a blockchain system. It quantifies the effort miners spend to solve cryptographic puzzles, influencing both the security and the distribution of the corresponding rewards. The higher the associated computational power, the more secure the blockchain becomes, as it would require a ill-intentioned hacker significant resources to compromise the system.⁹ However, this increase in computational power has made mining more challenging, requiring greater resources for a diminishing reward. The relationship between cryptocurrency valuation and computational power is addressed by Pagnotta and Buraschi (2018); Chiu and Koeppl (2022), who present theoretical models linking hashpower to network security, showing how mining effort reduces attack risk and stabilizes prices. In line with this, Prat and Walter (2021) point out that while the hash rate correlates with Bitcoin prices, the implied elasticity is not one as miners' investments create stability. Pagnotta (2022) finds that the prices of Bitcoin, Ethereum, and Litecoin are positively related to their hashrates.

The size of a blockchain network is the other key fundamental factor, typically measured by the number of daily active users. Active users are the unique IP addresses that have engaged with the blockchain in the past 24 hours, whether through transactions, mining, staking, or other activities. A larger number of active users generally correlates with higher liquidity and enhanced resilience to attacks or failures due to the decentralized

⁸Other algorithms, such as Proof of Authority (PoA), are less common but have potential applications in areas like central banking (Dziembowski et al., 2015).

⁹Bitcoin's hashrate, for example, is currently around 133 EH/s and this implies a massive computational effort (Noda et al., 2020). One EH/s equals 1,000,000,000,000,000,000 or one quintillion hashes per second.

nature of the system. The size of the network also serves as an indicator of the blockchain's overall adoption and health, as a higher number of active users reflects broader engagement, increased trust in the platform, and a stronger ecosystem supporting its ongoing functionality.¹⁰

Investor sentiment, often driven by "noise traders" who are presumed to be more influenced by market trends and recent price changes than by the underlying fundamentals, represents another key factor in cryptocurrency markets, see, e.g., [Jo et al. \(2020\)](#), [Mokni et al. \(2022\)](#), [Li et al. \(2022\)](#), and [Sakariyahu et al. \(2024\)](#). One way to gauge investor sentiment is through Google Trends, which tracks search activity and can provide insights into public interest in specific cryptocurrencies. Studies, such as [Ginsberg et al. \(2009\)](#) and [Guidolin et al. \(2018\)](#), have shown that Google Trends data can be an effective predictor in various asset markets, including cryptocurrencies.

Despite the theoretical advancements described above, there has been limited empirical validation of these insights, leaving a significant gap in the asset pricing literature. Our work seeks to bridge this gap by offering an empirical counterpart to theoretical models that highlight the critical roles of hashrate, network size, and sentiment in the valuation of digital assets. Building on the papers surveyed in this section, we identify these three factors as key drivers in our empirical analysis.

3 Data and Methodology

In the first part of the analysis, we focus on the construction of relevant factors and perform an in-sample evaluation of their explanatory power for cryptocurrency returns. Our data cover the period August 21, 2015 - September 30, 2022 and include 372 weekly observations. We use the daily prices to compute weekly returns by cumulating the daily returns of 7-day periods ending on Fridays. We use weekly returns to mitigate any day-of-the-week effects (see, e.g., [Biais et al. \(2023\)](#)) and problems due to the appearance of outliers. To build the factors we are after, we select five major cryptocurrencies—Bitcoin (BTC), Ethereum (ETH), Litecoin (LTC), Monero (XMR), and Dash (DASH)—which together represent a substantial portion of the overall cryptocurrency market, accounting

¹⁰[Cong et al. \(2021\)](#) develop a theoretical model of network externalities, in which a larger user base amplifies transactional benefits and induce equilibrium multiplicity. [Sockin and Xiong \(2023\)](#) provide a theoretical framework for cryptocurrency valuation based on positive network externalities, emphasizing decentralized control through tokenization.

for approximately between 70% to 90% of the total market capitalization over our sample. These cryptocurrencies were specifically chosen because their blockchain operations rely on miners for transaction validation and network functionality. To compute (log-) returns, similarly to [Bhambhwani et al. \(2023\)](#), we sourced weekly prices from Coin Metrics, which aggregates and weights prices from the most liquid cryptocurrency exchanges worldwide, with data derived from 22 exchanges worldwide. Cryptocurrency prices are reported in U.S. dollars (USD) and are recorded at 00:00 UTC on each day.

As discussed in Section 2, two fundamental blockchain metrics are used to capture key characteristics of the cryptocurrencies in our study: computational power and active user count. Computational power, measured in terahashes, reflects the processing power used by miners to secure the blockchain, while network size, measured by the number of unique active users (crucially, Coin Metrics does not double count active addresses with multiple transactions on a given day), provides insights into the scale of network participation. In the case of Monero (XMR), however, retrieving user data was not possible due to the privacy-focused design of its blockchain, which obfuscates the users' IP addresses ([Narayanan, 2016](#)). As for the measurement of sentiment, we resort to Google Trends data, which on a standardized 0-100 scale, reflect the relative search interest for each cryptocurrency. This search data, specifically focused on cryptocurrency names and their tickers applied to the entire world, serves as a proxy for investor sentiment. We calculated weekly growth rates for sentiment, using the same methodology applied to price data.

In the final stage of the analysis, we used benchmark weekly return data from the Coin Metrics Bletchley Index (CMBI) set to assess the significance of the average excess abnormal returns (alpha) derived from the managed portfolios that we shall build based on our forecast-sorting methodology. The primary benchmark is the CMBI10 index, which tracks the overall performance of the top ten cryptocurrencies by market capitalization. Additionally, we include the CMBIM index, a momentum-based factor derived from the top ten cryptocurrencies by market capitalization. Both benchmarks are also used to evaluate the robustness and forecast accuracy of our empirical models.

In terms of methodology, our analysis is divided into two key stages. The first step involves an in-sample analysis, through which we outline the construction of the factors and validate our hypotheses on the role of fundamentals and sentiment variables. The second step extends to an analysis, where we develop portfolios aimed at exploiting the predictability of cryptocurrency returns. In the case of the OOS analysis, we extend our

selection to a total of 41 cryptocurrencies, whose list is provided in Table 1. These represent the 41 biggest PoW cryptocurrencies in terms of average daily market capitalization during our sample period.¹¹

Table 1: List of cryptocurrencies used in the analysis

Ticker	Name	Mean	Standard Deviation
ADA	Cardano	0.94	0.15
ANT	Aragon	0.25	0.15
BAT	Basic Attention Token	0.42	0.13
BTC	Bitcoin	0.62	0.09
BTG	Bitcoin Gold	-0.68	0.15
CVC	Civic	0.21	0.16
DASH	Dash	0.83	0.13
DCR	Decred	0.28	0.11
DGB	DigiByte	0.03	0.15
DOGE	Dogecoin	1.79	0.19
DRGN	Dragonchain	-1.1	0.20
ELF	Aelf	0.16	0.16
ETC	Ethereum Classic	0.41	0.14
ETH	Ethereum	0.83	0.12
FUN	FunToken	0.42	0.16
GAS	GAS	0.51	0.15
GNO	Gnosis	0.19	0.12
GNT	Golem	0.14	0.15
KNC	Kyber Network Crystal	0.09	0.15
LEND	Aave	1.90	0.16
LINK	Chainlink	2.20	0.16
LTC	Litecoin	0.16	0.11
MAID	MaidSafeCoin	0.26	0.13
MANA	Decentraland	1.22	0.15
MKR	Maker	4.19	0.53
NEO	Neo	0.10	0.14
OMG	OMG Network	0.32	0.15
PAY	TenX	-2.15	0.15
POWR	Power Ledger	0.45	0.14
PPT	Populous	-1.43	0.16
REP	Augur	0.60	0.13
SNT	Status	0.19	0.15
VTX	Vertcoin	1.33	0.15
WAVES	Waves	0.39	0.17
WTC	Waltonchain	-1.28	0.13
XEM	NEM	0.81	0.13
XLM	Stellar	0.21	0.15
XRP	XRP	0.00	0.15
XVG	Verge	-0.99	0.16
ZEC	Zcash	-0.66	0.13
ZRX	0x	0.27	0.15

The table shows the 41 cryptocurrencies used in the out-of-sample analysis. The table includes the ticker, name, mean, and standard deviation of their weekly returns. The data cover the period January 5, 2018 - September 30, 2022.

¹¹Dash and Decred rely on a mix of Proof-of-Work and Proof-of-Stake (i.e., hybrid model) but we classify them as PoW for simplicity.

3.1 Factor Construction

Due to the non-tradable nature of blockchain characteristics, direct aggregation of such data to form an asset pricing factor is not feasible. As a result, these characteristics must be "projected" onto cryptocurrency returns. In fact, even if direct aggregation were possible, comparison would still be problematic because different blockchains employ distinct algorithms for calculating metrics such as computational power and active user counts. By projecting these blockchain characteristics onto cryptocurrency returns, we make these factors tradable in the market, thus enabling the empirical testing of procyclicality hypotheses (see [Barillas and Shanken \(2017\)](#)).

For the construction of the factors we follow the approach used by [Bhambhwani et al. \(2023\)](#), which was inspired by [Lettau and Ludvigson \(2001\)](#), who pioneered this methodology to estimate the impact of consumption growth rates on the stock market. Indeed, to define the factors, it is important to construct a factor-mimicking portfolio (FMP) for each characteristic. This process involves mapping the growth rate of each characteristic onto the returns of the test cryptocurrencies, effectively creating a portfolio that replicates as closely as possible (in a linear sense) the dynamics of the underlying factor. For instance, picking the case of the hash rate variable and of Ethereum (ETH) to illustrate our methodology, equation (1) describes the relationship between the computational power of ETH ($CP_{ETH,t}$) and the price changes of other cryptocurrencies:

$$\begin{aligned} \Delta CP_{ETH,t,t-\tau} = & \beta_1 \Delta Price_{BTC,t,t-\tau} + \beta_2 \Delta Price_{XMR,t,t-\tau} + \\ & + \beta_3 \Delta Price_{LTC,t,t-\tau} + \beta_4 \Delta Price_{DASH,t,t-\tau} \end{aligned} \quad (1)$$

Here, the return of the analyzed cryptocurrency (ETH's computational power) is omitted to avoid spurious effects. After obtaining the exposures of ETH hash rate to the price changes of each of the other factor-driving cryptocurrencies, the estimated coefficients are turned into weights by standardization, as shown below:

$$\begin{aligned}
W_{BTC} &= \frac{\beta_1}{\beta_1 + \beta_2 + \beta_3 + \beta_4} \\
W_{XMR} &= \frac{\beta_2}{\beta_1 + \beta_2 + \beta_3 + \beta_4} \\
W_{LTC} &= \frac{\beta_3}{\beta_1 + \beta_2 + \beta_3 + \beta_4} \\
W_{DASH} &= \frac{\beta_4}{\beta_1 + \beta_2 + \beta_3 + \beta_4} = 1 - W_{BTC} - W_{XMR} - W_{LTC}.
\end{aligned}$$

The FMP for ETH's computational power is then created by applying these weights to the price changes of the other cryptocurrencies, as shown in equation (3):

$$\begin{aligned}
FMP_{CP_{ETH}} &= W_{BTC} \times \Delta Price_{BTC,t,t-\tau} + W_{XMR} \times \Delta Price_{XMR,t,t-\tau} + \\
&+ W_{LTC} \times \Delta Price_{LTC,t,t-\tau} + W_{DASH} \times \Delta Price_{DASH,t,t-\tau}
\end{aligned} \tag{3}$$

We repeat this process for all other cryptocurrencies, generating an FMP for each characteristic (computational power, users, and Google Trends) for each asset. Once all FMPs for computational power are obtained, the aggregate factor is derived as shown in equation (4):

$$\begin{aligned}
HashrateF_t &= w_{BTC,t} \times FMP_{CP_{BTC,t}} + w_{XMR,t} \times FMP_{CP_{XMR,t}} + \\
&+ w_{LTC,t} \times FMP_{CP_{LTC,t}} + w_{DASH,t} \times FMP_{CP_{DASH,t}}
\end{aligned} \tag{4}$$

In this equation, the weights $w_{BTC,t}, w_{XMR,t}, w_{LTC,t}, w_{DASH,t}$ are determined by the relative market capitalizations of the respective cryptocurrencies. The same process is applied to the other characteristics, with Bitcoin's blockchain data excluded from factors unrelated to BTC. This approach allows us to evaluate whether the results remain consistent and robust even when the significant influence of Bitcoin, which dominates the cryptocurrency market, is removed from the analysis.

3.1.1 Pro-cyclicality of Factors

To establish the pro-cyclicality of the factors, we draw on the framework provided by [Cochrane \(2005\)](#), who defines pro-cyclical factors as those that are tradable in the market and that are associated with a positive risk premium. A positive risk premium indicates that these factors tend to generate higher returns when market conditions are favorable and marginal utility is low, while yielding lower returns during unfavorable market condi-

tions when marginal utility is high. Moreover, assets with a positive risk premium should exhibit a positive exposure to pro-cyclical factors, further reinforcing their alignment with favorable market conditions.

Building on these hypotheses, an initial test examines whether the factors exhibit a positive risk premium, with their characteristics summarized in Tables 2 and 3.

Table 2: Correlations between cryptocurrency returns

	BTC	ETH	XMR	LTC	DASH
BTC	1.00	0.53	0.56	0.70	0.53
ETH	0.53	1.00	0.51	0.49	0.58
XMR	0.56	0.51	1.00	0.51	0.60
LTC	0.70	0.49	0.51	1.00	0.51
DASH	0.53	0.58	0.60	0.51	1.00

The table shows the correlations of the weekly returns of the cryptocurrencies used for the construction of the factors: Bitcoin (BTC), Ethereum (ETH), Monero (XMR), Litecoin (LTC), and Dash. The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations.

Table 3: Summary statistics of the factor mimicking portfolio returns

Factors	Mean	Standard Deviation	Sharpe Ratio
Momentum	5.72	18.63	0.31
User Factor	1.61	8.59	0.19
Hashrate Factor	1.68	10.73	0.16
Google Sentiment Factor	1.57	8.95	0.18
User Factor Ex-BTC	1.66	9.33	0.17
Hashrate Factor Ex-BTC	1.63	11.06	0.14
Google Sentiment Factor Ex-BTC	19.35	246.27	0.08

The table presents the descriptive statistics for the seven factors constructed in the paper. The sample mean, standard deviation, and Sharpe ratios are reported. The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations.

As one would expect, in Table 3 all pairs of cryptocurrencies are positively correlated, with coefficients ranging between 0.49 to 0.70. Even though this is not indicated, all pairwise Pearson correlation coefficients are statistically significant at standard levels. In Table 3, all the factor-mimicking portfolios imply a positive and relatively large weekly Sharpe ratio, ranging from 0.08 for the sentiment factor to the large 0.31 obtained for momentum. Visibly, the different factors imply heterogeneous total risk, as measured by their return standard deviation. This narrows the range of the Sharpe ratios to plausible levels.

To then study the exposure of cryptocurrency returns to the factors, we employ standard OLS regressions. The first regression does not involve the created factors as the cryptocurrency returns were simply regressed against the return of BTC and the momentum factor, as illustrated in equation (5):

$$Return_{Crypto,t} = \alpha + w_{BTC}Return_{BTC,t} + w_{MOM}MOM_t + \epsilon_t \quad (5)$$

In addition, we analyze another model in which each the fundamental factors developed earlier are individually incorporated into a classical three-factor model:

$$Return_{Crypto,t} = \alpha + w_{BTC}Return_{BTC,t} + w_{MOM}MOM_t + w_{Factor}Factor_t + \epsilon_t \quad (6)$$

Finally, given Bitcoin's central role in the cryptocurrency space, a robustness test was conducted by extending the three-factor models when the fundamental factors were constructed without including BTC returns (hereafter Ex-BTC Factors). The results of these tests are presented in Section 4.1 below.

3.2 Out-of-Sample Analysis

We now present the major contribution of this paper, stemming from the concept of utilizing factors to forecast future returns of crypto assets. We provide a concise overview of the predictive model and the statistical measures employed to evaluate its accuracy. Lastly, Section 3.4 outlines the methodology used for portfolio construction based on predictive performance.

A predictive regression model can be described by equation (7):

$$r_{t+1} = \alpha_i + \beta_i x_{i,t} + \epsilon_{t+1}, \quad (7)$$

where r_{t+1} denotes the return of the cryptocurrency at time $t + 1$, and $x_{i,t}$ represents one of the factors with which future returns of the crypto asset are predicted. It should be emphasized that the predictor is measured at time t , which is one period before the target variable. Following the approach in [Welch and Goyal \(2008\)](#), predictions are made for the following week (i.e., we compute one-step-ahead point forecasts). To estimate the parameters by OLS, the sample is progressively expanded to include the most recent data, adopting what represents an expanding window estimation approach.

The predictive accuracy is assessed relative to a benchmark commonly adopted in the literature, defined by the simple mean $\bar{r} = \sum_{j=1}^t r_j$.¹² A benchmark model is obtained

¹²The most widely used benchmarks in the literature are the simple historical average and the last available ("no-change in price") data point, which in this case would consist of a zero return forecast for returns. The former benchmark is more useful than the latter when analyzing a historical series that

when equation (7) collapses to become a regression only including the constant and with no predictors. Intuitively, if a model's prediction of future return $\hat{r}_{t+1}$ performs better than the benchmark $\bar{r}_{t+1}$, it can be inferred that $x_{i,t}$ contains valuable information for predicting the future return of a cryptocurrency. Sections 3.2.1 and 3.2.2 provide useful statistics for comparing the models against the benchmark.

3.2.1 Out-of-Sample R-Squared

The first metric which we employ to evaluate the predictive performance is the Out-of-Sample R^2 (R_{OS}^2), as defined in equation (8). This metric, originally brought to the attention of finance researchers by [Campbell and Thompson \(2008\)](#), is used to compare the predictions of the model, $\hat{r}_{t+1}$, with those of the benchmark, $\bar{r}_{t+1}$:

$$R_{OS}^2 = 1 - \frac{\sum_{k=q_0+1}^q (r_{m+k} - \hat{r}_{m+k})^2}{\sum_{k=q_0+1}^q (r_{m+k} - \bar{r}_{m+k})^2} \quad (8)$$

where R_{OS}^2 measures the mean squared forecast error of the model relative to that of the benchmark. When $R_{OS}^2 > 0$, the model outperforms the benchmark. To test the significance of the results, we rely on the statistics developed by [Diebold and Mariano \(2002\)](#) and [West \(1996\)](#).¹³ [Campbell and Thompson \(2008\)](#) demonstrated, within the context of the stock market, that even a marginally positive R_{OS}^2 —such as 0.5% for monthly data and 1% for quarterly data—can yield significant economic gains for a mean-variance investor. In our case based on weekly data, even smaller R_{OS}^2 s coefficients of 10-20 basis points may therefore be of economic relevance.

3.2.2 Certainty Equivalent Return

One of the main limitations of the R_{OS}^2 is its inability to account for an investor's risk aversion. To address this shortcoming, we introduce an additional metric to evaluate a models' validity: the certainty equivalent return (CER). The certainty equivalent is obtained by calculating the average utility of an portfolio manager who uses the factors to forecast the return of the crypto asset, and to compare it to the utility of an investor

are "mean-reverting," while the latter makes more sense when analyzing historical asset priceseries that follow a random walk.

¹³However, [Diebold and Mariano \(2002\)](#) and [West \(1996\)](#)'s statistics can lead to erroneous conclusions when comparing predictions from nested linear models leading the test to have very low power. Nonetheless, this case never occurs in our applications.

that is basing her decisions on the benchmark model, represented by the simple mean. The CER is defined as the sure, riskless rate of return that would make an investor indifferent between a given prediction model and the benchmark (or between two forecast models).

Specifically, we consider a reference agent with a simple, static mean-variance utility function and a risk aversion coefficient of three. This investor allocates wealth according to the mean-variance optimization framework, determining the optimal weight to allocate to the risky asset at time t , based on the return forecast for time $t + 1$ generated by their model, as described by equation (9):

$$w_{0,t} = \frac{1}{\gamma} - \frac{\bar{r}_{t+1}}{\hat{\sigma}_{t+1}^2} \quad (9)$$

where $\bar{r}_{t+1}$ represents the forecast obtained from the benchmark and $\hat{\sigma}_{t+1}^2$ is the sample variance estimated using rolling windows of 40 weeks, common to all models tested and compared and typical in the predictability literature (see [Rapach et al. \(2010\)](#)). At the end of each period, the investor will achieve an average utility equal to $\hat{v}_0$:

$$\hat{v}_0 = \hat{\mu}_0 - \frac{1}{2}\gamma\hat{\sigma}_0^2 \quad (10)$$

In equation (10), $\hat{\mu}_0$ and $\hat{\sigma}_0^2$ denote, respectively, the mean and variance of the portfolio constructed using the benchmark model as the predictor.

To calculate the certainty equivalent return, the steps outlined in equations (9) and (10) must be repeated for each of the models we examine. The only modification occurs in equation (9), where $\bar{r}_{t+1}$ is replaced by $\bar{r}_{t+1}$, as a different forecasting model is used in place of the benchmark model to predict the future return of the crypto asset. After performing these steps, we calculate the average utility for each model. The certainty equivalent is then determined as the difference between the average utility of the models and the average utility of the benchmark model. The CER can be interpreted as the management fee an investor is willing to pay to access a given forecasting model ([Marquering and Verbeek, 2004](#); [Campbell and Thompson, 2008](#); [Welch and Goyal, 2008](#); [Wachter and Warusawitharana, 2009](#)).

3.3 Portfolios and Significance Tests

As regards the construction of a static investment strategy, we aim to leverage on the heterogeneous predictability of alternative cryptocurrencies. Building on the tests conducted in previous analyses, cryptocurrencies can be classified and hence sorted in portfolios according to their forecasting accuracy. Specifically, a cryptocurrency with a higher R_{OS}^2 or certainty equivalent is indicative of more predictable future returns compared to others.

To assess whether any predictability uncovered earlier can generate economic value to an investor, over and beyond the results of the mean-variance asset allocation strategy pursued according to the steps explained earlier, we construct 16 portfolios from the full test set of 41 cryptocurrencies listed above. On each week of our time-period, we first rank all cryptocurrencies according to the R_{OS}^2 statistic, which is employed to evaluate their forecasting quality. This is separately applied to each of the predictive factor (mimicking) portfolio returns. Once the cryptocurrencies are ranked, we construct the portfolios by shorting those 20 expected to be less predictable than the median portfolio and go long in those 20 expected to be more predictable, resulting in a long-short portfolio with zero initial investment. This process is repeated afresh on every month on the basis of the cumulative forecasting performance of each model up to the point. Because each predictive factor originates a different sequence of sortings and hence long-short strategies, the OOS results shall depend and be informative on each of the factors taken in isolation from the others.

4 Empirical Results

In this section, we first present and analyze the results of the in-sample regressions. The transition from a two-factor model, which includes BTC and momentum (Table 4), to a three-factor model, where the fundamental factors are introduced individually (Table 5), is examined in detail.

In the second part, focused on the analysis, we present the results for the portfolio sorting of the 41 cryptocurrencies under study (Tables 7 and 8). This analysis evaluates the predictive performance of the models compared to the benchmark approach based on the simple average. Finally, Table 9 presents the results of the backtesting performed on the investment strategies that exploit the possibility to build long-short portfolios based on the relative back-tested predictability of cryptocurrencies vs. alternative factors.

4.1 In-Sample Results

Table 4: In-sample analysis results

	BTC	ETH	XMR	LTC	DASH
<i>Panel (a): 2-Factor Model</i>					
Alpha (%)	1.31 (2.62)	0.54 (0.79)	-0.39 (-0.59)	-0.94 (-1.80)	-0.18 (-0.27)
BTC Return	–	0.83 (10.84)	0.86 (11.75)	0.99 (17.13)	0.82 (10.99)
Momentum Factor	0.06 (2.23)	0.10 (2.94)	0.16 (4.65)	0.09 (3.50)	0.03 (0.97)
R-square	0.01	0.70	0.36	0.51	0.28
<i>Panel (b): 3-Factor Model Including the User Factor</i>					
Alpha (%)	0.45 (1.75)	0.80 (1.79)	-0.28 (-0.37)	-0.78 (-1.87)	0.04 (0.09)
BTC Return	–	-0.92 (-9.38)	-0.01 (-4.44)	-0.05 (-0.57)	-0.72 (-6.76)
Momentum Factor	-0.04 (-2.93)	-0.02 (-1.03)	0.06 (2.07)	0.02 (0.76)	-0.08 (-3.08)
User Factor	0.88 (30.04)	2.07 (20.73)	13.97 (20.73)	1.24 (13.26)	1.83 (16.76)
R-square	0.74	0.70	0.60	0.68	0.62
<i>Panel (c): 3-Factor Model Including the Hashrate Factor</i>					
Alpha (%)	0.76 (2.24)	0.86 (1.76)	-0.12 (-0.23)	-1.71 (-1.80)	0.18 (0.48)
BTC Returns	–	-0.22 (-2.74)	-0.04 (-0.46)	0.37 (5.14)	-0.38 (-5.95)
Momentum Factor	-0.02 (-1.33)	0.01 (0.36)	0.08 (2.82)	0.04 (1.65)	-0.07 (-3.68)
Hashrate Factor	0.61 (19.62)	1.16 (17.41)	0.99 (14.19)	0.68 (11.49)	0.96 (15.93)
R-square	0.60	0.72	0.66	0.66	0.66
<i>Panel (d): 3-Factor Model Including the User and Hashrate Factors</i>					
Alpha (%)	0.29 (2.36)	1.16 (1.93)	0.19 (0.33)	-0.19 (-0.60)	0.56 (1.07)
BTC Returns	–	-1.88 (-6.89)	1.73 (6.55)	-2.33 (-16.44)	-2.49 (-10.36)
Momentum Factor	-0.02 (-3.29)	0.03 (1.08)	0.09 (3.00)	0.01 (0.49)	-0.05 (-1.89)
Google Sentiment Factor	0.94 (70.91)	2.70 (10.22)	2.58 (10.13)	3.31 (24.18)	3.29 (14.19)
R-square	0.94	0.70	0.51	0.83	0.56

The table presents the estimates of the regressions of weekly series of cryptocurrency returns on two- and three-factor models. The cryptocurrencies analyzed are: Bitcoin (BTC), Ethereum (ETH), Monero (XMR), Litecoin (LTC), and DASH. Panel (a) shows the results of a two-factor model regression, which includes the return of BTC and the momentum factor. Panels (b)-(d) report the estimates of three-factor regressions obtained by adding the User (b), the Hashrate (c), and the Google (d) factor mimicking portfolios. The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations. HAC t-statistics are reported in parentheses. The R-squared coefficients are boldfaced when the corresponding F-test of overall model significance rejects the null hypothesis.

In the case of the two-factor model, in which past Bitcoin returns and the returns on the momentum factor mimicking portfolio explain the returns of the four baseline cryptocurrencies, each of them exhibits a positive, significant exposure to Bitcoin's return, with coefficients ranging from 0.82 in the case of Dash to 0.99 in the case of Litecoin.

However, Panel A of Table 4 shows that the estimated exposure to the momentum factor is much lower, with coefficients ranging from 0.03 for Dash to 0.10 for Ethereum, even though four out of five coefficients are statistically significant. The R^2 of these regressions implies an overall explanatory power that goes from a minimum of 0.28 for Dash to a maximum of 0.7 for Ethereum. The table suggests that Bitcoin returns serve as a meaningful factor in describing cryptocurrency returns, while the findings concerning the momentum factor aligns with those in [Liu and Tsyvinski \(2021\)](#).

Panels B, C, and D of Table 4 show results of the three-factor models, in which cryptocurrency returns are regressed on Bitcoin, momentum, and one of the three fundamental factor mimicking portfolio returns (Users, Hashrate, and Google, taken one at the time). Broadly similar results emerge across these richer models: all cryptocurrency return series display positive, precisely estimated exposures to each fundamental factor. Additionally, the sign of the estimated loading on Bitcoin returns shifts across models: in the two-factor model, such an estimated beta turns out to be positive across all cryptocurrencies, while in the three-factor models, it becomes negative and it is generally statistically significant. For example, Panel B shows negative BTC exposures for all cryptocurrency return series, ranging from -0.92 for ETH to -0.01 for LTC (and both are precisely estimated). Comparing R^2 coefficients across models, reveals much higher explanatory power to lie with the three-factor model than in the two-factor model. The only exception is when using the Google factor for ETH, where the R^2 is essentially unchanged. Table 5 integrates the information in Table 4 by also showing the estimated pairwise correlations among the factor mimicking portfolios used in this paper. Here, BTC is found to be strongly correlated with the fundamental factors, ranging from 0.97 in the case of the Google factor to 0.75 for the Hashrate factor, possibly explaining why BTC sensitivity coefficients change signs in the three-factor models vs. the two-factor one, possibly as a result of the considerable collinearity characterizing BTC returns relative to the remaining factors.¹⁴

We test the robustness of these in-sample results by building three-factor models that use versions of the factor-mimicking portfolios that exclude bitcoin from their construction, to allow BTC to play a role only through its returns to be included in the regressions. Table 6 reports the corresponding results, which align closely with those in Table 4, except

¹⁴A few additional pairs of factors are characterized by large correlations, for instance the User and HashRate and sentiment factors, but it is then crucial to note that the returns on these factor mimicking portfolios are never used jointly in the prediction models. Of course, one expects to find large correlations between factor returns computed using all cryptocurrencies and those ex-bitcoin data and, even though this is unsurprising, we note a few instances of small and statistically insignificant correlations.

Table 5: Pairwise correlations between factors

	Bitcoin	Momentum	User Factor	HashRate Factor	Google Factor	User Factor Ex-BTC	HashRate Factor Ex-BTC	Google Factor Ex-BTC
Bitcoin	1	0.12	0.86	0.76	0.97	0.69	0.67	0.14
Momentum	0.12	1	0.24	0.23	0.17	0.24	0.23	-0.06
User Factor	0.86	0.24	1	0.97	0.94	0.95	0.91	0.00
HashRate Factor	0.76	0.23	0.97	1	0.87	0.98	0.96	-0.07
Google Sentiment Factor	0.97	0.17	0.94	0.87	1	0.81	0.80	0.16
User Factor Ex-BTC	0.69	0.24	0.95	0.98	0.81	1	0.97	0.21
HashRate Factor Ex-BTC	0.67	0.23	0.91	0.96	0.80	0.97	1	-0.17
Google Sentiment Factor Ex-BTC	0.14	-0.06	0.00	-0.07	0.16	0.21	-0.17	1

The table shows the pairwise sample correlations of the weekly returns of the factors used in the in-sample regression analysis: Bitcoin, Momentum, User, Hashrate, User Ex-BTC, Hashrate Ex-BTC, Google Ex-BTC. The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations. Significant coefficients based on a 5% size test are boldfaced.

for Panel C, where the Google Factor Ex-BTC is used. Here, in contrast to earlier results, the exposure of cryptocurrency returns to this sentiment factor turns negative, while the loading on BTC returns remains positive. This difference suggests that excluding Bitcoin returns from the construction of a sentiment factor can lead to skewed findings, reflecting Bitcoin’s central role and significant influence on the overall cryptocurrency market.

4.2 Out-of-Sample Results

Our OOS tests examine predictive accuracy by comparing whether the factors we have created in Section 3 can forecast cryptocurrency returns more effectively than a simple average does. Beginning with the model in equation (5), a battery of statistics (e.g., the OOS R^2 , the CER, the cumulative forecast error) are calculated to contrast these results with a benchmark, simple historical average for cryptocurrency returns. Results are organized in two displays: Table 7 presents results for the raw fundamental factors while Table 8 shows our findings when the Ex-BTC factors are used. Because we use each of them in isolation from the others, we present the main findings in a separate panel, one per each factor. Figures 4 and 5 visually represent prediction accuracy across models for four selected cryptocurrencies. Their goal is to show the difference between the prediction errors of each three-factor predictive regression and that of a benchmark model which forecasts future cryptocurrency returns using the historical mean (Welch and Goyal, 2008).¹⁵ In the figures, the first row of four plots concern the prediction errors derived from using raw fundamentals and momentum to forecast cryptocurrency returns,

¹⁵The cryptocurrencies used in these plots are BTC, ETH, LTC, and DASH, as these yield the longest available historical time series. To compute the cumulative forecast error, net of an initialization sample composed of 104 observations given the simple nature of our regression models, the OOS period comprised 268 weekly data points between October 2017 and September 2022.

Table 6: In-sample analysis results when BTC is excluded from the construction of factor mimicking portfolios

	BTC	ETH	XMR	LTC	DASH
<i>Panel (a): 3-Factor Model with User Factor Ex-BTC</i>					
Alpha (%)	0.76 (2.06)	0.66 (1.47)	-0.28 (-0.61)	-0.89 (-1.86)	-0.06 (-0.15)
BTC Beta		-0.13 (-1.91)	-0.01 (-0.18)	0.60 (8.28)	-0.14 (-2.10)
Momentum Beta	-0.02 (-1.11)	-0.00 (-0.14)	0.06 (2.47)	0.05 (1.99)	-0.07 (-3.27)
Factor User Beta	0.64 (16.61)	1.32 (20.45)	1.21 (18.21)	0.54 (7.84)	1.32 (21.50)
R-square	0.47	0.70	0.69	0.59	0.71
<i>Panel (b): 3-Factor Model with Hashrate Factor Ex-BTC</i>					
Alpha (%)	0.90 (2.41)	0.78 (1.30)	-0.01 (-0.23)	-0.80 (-1.65)	0.21 (0.57)
BTC Beta		0.23 (2.59)	-0.11 (-2.04)	0.65 (8.96)	-0.17 (-3.10)
Momentum Beta	-0.02 (-0.78)	0.04 (1.30)	0.05 (2.84)	0.06 (2.26)	-0.07 (-3.67)
Hashrate Beta	0.51 (15.95)	0.68 (9.93)	1.11 (26.58)	0.39 (7.06)	1.12 (26.72)
R-square	0.45	0.46	0.80	0.57	0.78
<i>Panel (c): 3-Factor Model with Google Factor Ex-BTC</i>					
Alpha (%)	1.39 (2.80)	0.02 (0.03)	-0.82 (-1.38)	-0.48 (-1.17)	-0.45 (-0.70)
BTC Beta		0.94 (14.39)	0.96 (14.39)	0.90 (19.27)	0.88 (12.12)
Momentum Beta	0.06 (2.42)	0.08 (2.56)	0.14 (4.46)	0.12 (5.51)	0.02 (0.58)
Google Sentiment Factor Beta	0.01 (2.76)	-0.03 (-11.21)	-0.02 (-9.01)	0.02 (13.71)	-0.01 (-5.26)
R-square	0.03	0.50	0.49	0.69	0.34

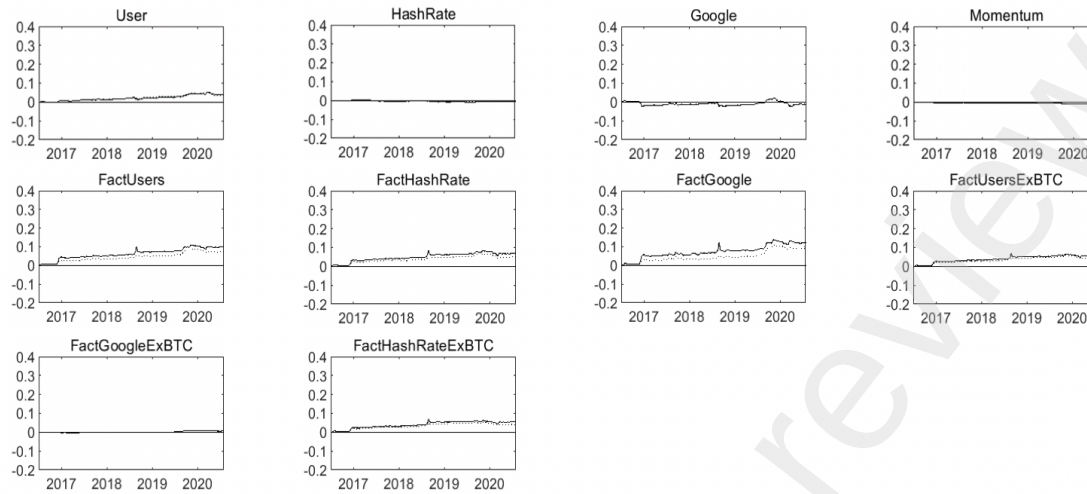
The table presents the estimates of the regressions of weekly series of cryptocurrency returns on two- and three-factor models, when the factors are constructed excluding BTC returns data. The cryptocurrencies analyzed are: Bitcoin (BTC), Ethereum (ETH), Monero (XMR), Litecoin (LTC), and DASH. Panel (a) shows the results of a two-factor model regression, which includes the return of BTC and the momentum factor. Panels (b)-(d) report the estimates of three-factor regressions obtained by adding the User (b), the Hashrate (c), and the Google (d) factor mimicking portfolios. The sample covers the period August 21, 2015 - September 30, 2022 and includes 372 weekly observations. HAC t-statistics are reported in parentheses. The R-squared coefficients are boldfaced when the corresponding F-test of overall model significance rejects the null hypothesis.

while the remaining six plots concern the forecasting power of factor mimicking portfolios formed on the basis of Users, HashRate, and Google sentiment, built either using BTC data or omitting BTC.¹⁶

In the plots, when the predictive regression model outperforms the benchmark, the curve lies above the (zero) horizontal axis; when the curve rises, irrespective of whether the curve is positive or negative, then the predictive factor regression model shows an

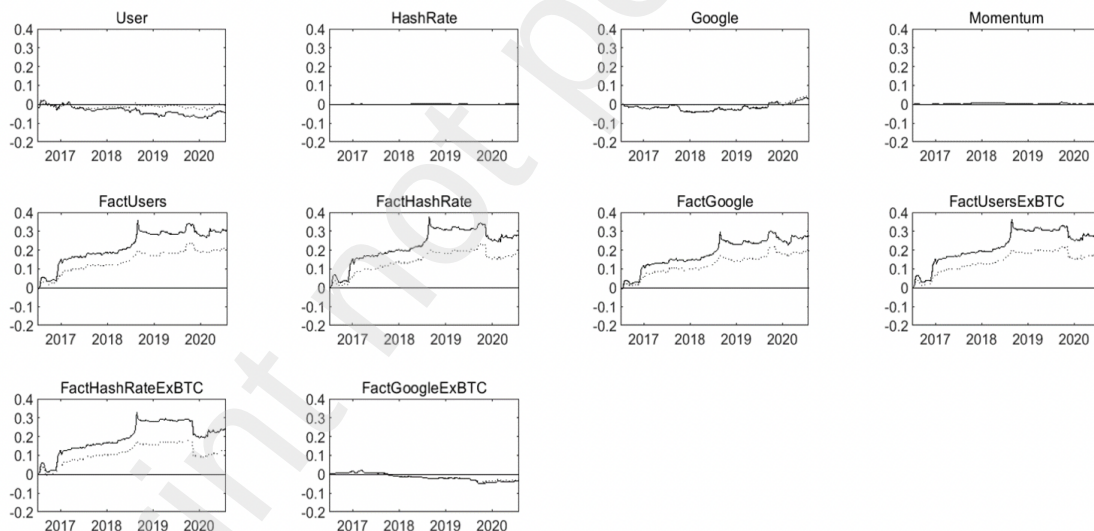
¹⁶The momentum factor is by definition a mimicking portfolio.

Figure 4
Cumulative prediction errors



(a) Bitcoin

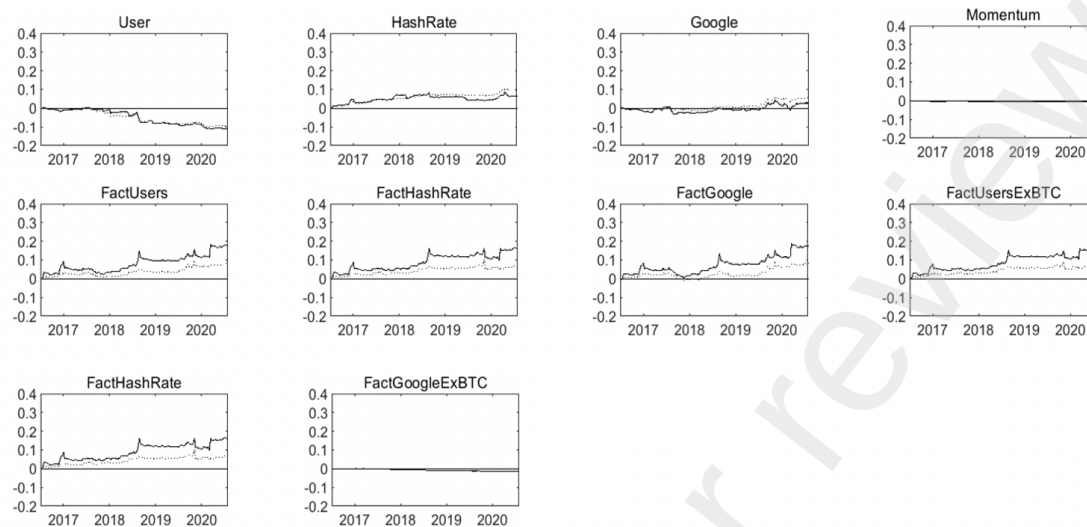
With reference to Bitcoin returns, the charts represent the difference between the prediction error of the historical mean and the prediction error of a predictive factor regression that uses as a forecast instrument the variable indicated in correspondence to the plot. The dashed line represents Campbell and Thompson's adjustment. The sample covers the period October 2017 - September 2022 and is based on a total of 268 weekly observations.



(b) Ethereum

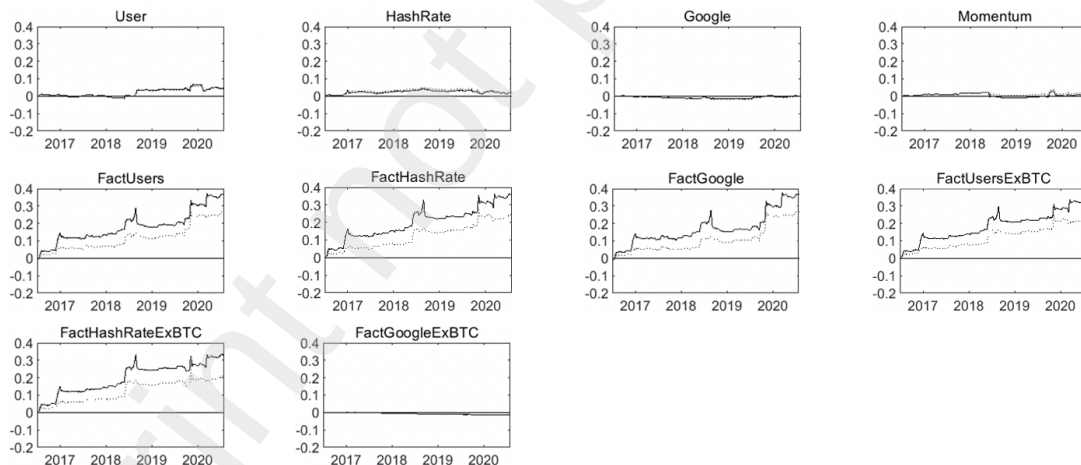
With reference to Ethereum returns, the charts represent the difference between the prediction error of the historical mean and the prediction error of a predictive factor regression that uses as a forecast instrument the variable indicated in correspondence to the plot. The dashed line represents Campbell and Thompson's adjustment. The sample covers the period October 2017 - September 2022 and is based on a total of 268 weekly observations.

Figure 5
Cumulative prediction errors



(a) Litecoin

With reference to Litecoin returns, the charts represent the difference between the prediction error of the historical mean and the prediction error of a predictive factor regression that uses as a forecast instrument the variable indicated in correspondence to the plot. The dashed line represents Campbell and Thompson's adjustment. The sample covers the period October 2017 - September 2022 and is based on a total of 268 weekly observations.



(b) Dash

With reference to Dash returns, the charts represent the difference between the prediction error of the historical mean and the prediction error of a predictive factor regression that uses as a forecast instrument the variable indicated in correspondence to the plot. The dashed line represents Campbell and Thompson's adjustment. The sample covers the period October 2017 - September 2022 and is based on a total of 268 weekly observations.

improvement relative to benchmark. Of course, when the curves lie below the horizontal axis, this means that a predictive factor model underperforms the historical sample mean and the fact the curve declines implies that the former is getting worse vs. the latter. Besides the cumulative prediction error differences, the plots report (as a dashed line) such a difference after applying the simple correction proposed by [Campbell and Thompson \(2008\)](#) to prevent predictive models from generating economically nonsensical forecasts, such as negative expected returns in a context where it is unreasonable.¹⁷

In Figures 4 and 5, the four panels reveal that most predictive models based on fundamental factor mimicking portfolios outperform the simple historical average. Moreover, the factor mimicking portfolios do visibly better than the raw factors in the first row, while momentum never shows any appreciable forecasting power on top of the benchmark. In fact, in the case of ETH and LTC, the raw User factor yields a negative and worsening relative cumulative predictive accuracy performance while the corresponding mimicking factor performs very well already starting in 2016. The same occurs for the HashRate in the case of BTC and ETH and for the Google sentiment factor in the case of LTC and DASH. The increased effectiveness of fundamental characteristics when they are represented by composite mimicking factors may relate to [Rapach et al. \(2010\)](#)'s findings, who demonstrated that combining predictions reduces forecast volatility, yielding better outcomes in an equity context.¹⁸

Finally, all plots show that while building the predictive mimicking portfolios for Users and the HashRate using or not using BTC data hardly makes any difference across all cryptocurrencies. As expected in the light of the findings in Table 6, where computing the sentiment factor-mimicking risk driver without accounting for the role played by BTC was leading to substantial biases, the predictive power of the Google sentiment portfolios vanishes altogether when it is defined excluding the information brought about by BTC

¹⁷They suggest to impose constraints on the predicted returns to ensure economic plausibility, in our case that expected cryptocurrency returns cannot be negative. Note that in our case, one can find rational reasons for expected cryptocurrency returns to turn negative, but given their remarkable level of risk that would still be rather unusual. Hence, even though less than compelling vs. [Campbell and Thompson \(2008\)](#)'s application to excess equity returns, we felt it would be prudent to also report it among our results.

¹⁸[Rapach et al. \(2010\)](#) combined forecasts ex post; thus, predictions were estimated for each model and subsequently combined, all models sharing the objective of predicting the equity market risk premium. In this paper, however, factors are calculated by combining fundamental characteristics across various cryptocurrencies ex ante. This approach results in reduced volatility for the factors compared to individual fundamental characteristics, which, in turn, leads to a reduction in the volatility of the forecast itself, $\hat{r}_{t+1}$.

and, apart from the case of BTC itself (oddly enough), leads to forecasting performances that are worse than the benchmark for the remaining currencies.

Visibly, in Figures 4 and 5, by the end of our OOS period, most of the factor mimicking portfolio predictors provide a cumulative spread in the order of 10-35 percent (depending on whether or not the Campbell and Thompson's correction is applied) for all cryptocurrencies. The largest cumulative improvements vs. the historical mean are brought forth for DASH (when the HashRate factor scores an almost 40 percent improvement) and the smallest in the case of Bitcoin. In the latter case, the sentiment Google factor mimicking portfolio yields the highest cumulative predictive improvement, approximately 15 percent by the end of our OOS period (10 percent when negative forecasts are floored at zero). In light of the fact that BTC (DASH) is the (most recently created) longest standing cryptocurrencies, with the deepest (thinnest) and most (less) efficient set of trading markets, the reported distribution of results across cryptocurrencies may be sensible. Finally, in terms of dynamics over time, all cryptocurrencies show an increasingly strong forecasting power of the fundamental factor mimicking factors, with an acceleration and local peak in late 2018 followed by a further climb up in 2021-2022 (this is the case of DASH and LTC) and a peak-and-shoulders pattern between 2018 and 2019 followed by further gains in 2021 in the case of ETH.

After the visual intuition offered by Figures 4 and 5, we proceed to analyze the findings presented in Tables 7 and 8 for our full cross-section of 41 cryptocurrencies.¹⁹

In the two tables, also to save space vs. the case in which would report the differential cumulative predictive performances over time, the summary indicator of forecasting power that appears is the OOS R^2 (R_{OS}^2), with the associated p-value calculated using the method of Diebold and Mariano (2002).²⁰ Table 7 predominantly displays positive R_{OS}^2 values, indicating that for most factor mimicking predictors, the corresponding predictive regression generally provides stronger OOS precision than a simple average. In Table 7, all fundamental factors are equally successful (the null of a non positive R_{OS}^2 can be

¹⁹Here we need to emphasize that the sample expands to 41 cryptocurrencies in the cross-sectional dimension at the expense of the length of the time series, which is restricted to a January 2018 - September 2022 sample. The 41 cryptocurrencies are derived from an initial set of the top 100 cryptocurrencies by trading volume, by dropping those for which the HashRate fundamental factor would not make sense because they use PoS as a consensus mechanism, which would not allow for the estimation of a hashrate. The estimation period for the model parameters remains however 104 weeks (i.e., 2 years) and hence the OOS period spans January 2020 - September 2022, for a total of 140 observations.

²⁰The null hypothesis $R_{OS}^2 \leq 0$ is tested against the alternative hypothesis $R_{OS}^2 > 0$. using a standard t-ratio test statistic, in which the estimated standard error is HAC-adjusted

Table 7: OOS predictive results for a cross-section of cryptocurrencies

Ticker	User Factor			Hashrate Factor			Google Factor		
	OOS R-square	p-value	CER ($\gamma = 3$)	OOS R-square	p-value	CER ($\gamma = 3$)	OOS R-square	p-value	CER ($\gamma = 3$)
ADA	10.88	0.00	5.85	11.68	0.00	6.61	9.73	0.00	5.22
ANT	1.50	0.16	0.76	0.49	0.22	-0.06	1.71	0.14	1.10
BAT	5.35	0.03	2.35	4.17	0.04	2.46	5.86	0.03	2.36
BTC	6.27	0.01	3.01	3.81	0.02	2.20	7.84	0.01	3.83
BTG	11.41	0.00	2.09	10.82	0.00	0.73	11.15	0.00	2.11
CVC	2.89	0.10	0.58	1.52	0.09	0.29	3.35	0.09	0.94
DASH	9.75	0.01	0.14	9.20	0.01	-0.09	9.95	0.01	-0.01
DCR	7.82	0.01	4.93	6.09	0.01	3.73	9.07	0.01	3.56
DGB	9.91	0.00	2.32	9.08	0.00	2.14	9.97	0.00	2.33
DOGE	2.17	0.06	0.57	2.57	0.04	2.16	1.51	0.11	0.92
DRGN	1.84	0.07	0.41	0.34	0.17	1.31	2.31	0.06	0.00
ELF	4.36	0.01	1.61	4.69	0.01	1.98	3.67	0.02	1.24
ETC	6.63	0.01	2.09	6.46	0.01	1.61	5.89	0.01	2.05
ETH	10.06	0.01	5.94	8.97	0.00	5.95	8.97	0.00	5.56
FUN	0.50	0.22	0.81	0.32	0.28	0.23	0.98	0.22	0.49
GAS	10.40	0.01	0.61	12.92	0.02	0.19	8.34	0.01	1.71
GNO	6.17	0.00	3.99	5.72	0.00	3.83	5.08	0.00	3.69
GNT	4.75	0.02	1.79	5.53	0.03	1.64	3.87	0.03	1.57
KNC	0.85	0.13	0.84	0.22	0.16	0.95	0.44	0.14	0.57
LEND	0.15	0.18	0.52	0.18	0.18	0.45	1.08	0.27	0.16
LINK	33.19	0.00	15.60	34.77	0.00	16.45	27.34	0.00	14.87
LTC	8.64	0.01	2.79	7.68	0.01	2.39	7.41	0.01	1.98
MAID	1.46	0.39	-1.01	2.06	0.44	-1.58	1.64	0.42	-1.38
MANA	3.56	0.03	1.86	3.82	0.05	1.71	2.82	0.05	1.63
MKR	15.35	0.00	12.80	14.22	0.00	14.02	14.07	0.00	14.07
NEO	14.33	0.00	5.56	12.66	0.00	4.79	12.74	0.00	4.94
OMG	2.91	0.02	1.06	1.56	0.04	0.27	1.93	0.04	0.25
PAY	7.75	0.05	0.14	2.26	0.06	0.02	1.73	0.08	0.03
POWR	3.69	0.04	0.91	3.45	0.04	0.26	3.89	0.04	0.17
PPT	0.09	0.32	0.23	0.58	0.38	0.09	0.55	0.37	0.08
REP	3.17	0.06	0.25	2.87	0.06	0.08	2.72	0.07	0.19
SNT	2.67	0.10	1.82	1.40	0.12	1.30	1.14	0.13	0.99
VTC	1.98	0.13	0.62	2.11	0.11	0.12	2.15	0.12	0.32
WAVES	5.23	0.02	0.62	3.97	0.02	0.68	4.00	0.01	0.73
WTC	7.71	0.02	-2.19	6.08	0.03	2.45	5.83	0.03	2.00
XEM	3.29	0.03	0.02	4.42	0.02	0.26	4.68	0.02	0.97
XLM	9.46	0.01	2.39	8.26	0.01	2.63	7.52	0.01	2.80
XRP	9.98	0.07	0.28	3.47	0.04	0.65	3.47	0.04	0.92
XVG	6.78	0.01	1.17	5.60	0.01	2.22	5.60	0.02	0.94
ZEC	10.42	0.01	0.99	9.09	0.01	1.49	8.41	0.01	1.56
ZRX	5.30	0.01	2.34	4.02	0.02	2.27	4.07	0.02	2.08
Average	6.60	0.04	2.18	5.83	0.05	2.22	5.72	0.06	2.18

The table presents the OOS summary predictive performance results of the three factor-mimicking portfolio predictors isolated earlier and applied to 41 cryptocurrencies. The table reports the cryptocurrency ticker, the OOS predictive percentage R-square, the associated p-value, and the annualized, percentage certainty equivalent for a mean-variance investor characterized by a risk aversion coefficient of 3. The OOS R-squared that are boldfaced are significant with a p-value of 5% or less.

rejected 29, 28, and 27 times out of 41, respectively), but on average rather successful as in 60-70 percent of the cases the R_{OS}^2 turns out to be significant while under the null this should occur only 5 percent of the time. Interestingly, although the edge is rather small (29 and 28 vs. 27) the User and HashRate fundamentals are better predictors than Google sentiment turns out to be.²¹ It is also notable that when one predictor tends to

²¹However, if one were to average out only the statistically significant R_{OS}^2 statistics, this would give a small advantage to the sentiment predictor.

work for a given ticker, this extends throughout to all predictors. This means that it is the time series of cryptocurrency returns that contains pockets of predictability and this may be picked up both by fundamental and by sentiment-driven forecasting. In the case of the CER results, visibly these are overwhelmingly positive throughout (in 38 tickers in the case of the fundamental predictors and 39 in the case of Google sentiment), for all tickers and predictors, i.e., factor mimicking-based predictability is never wealth-destroying in risk-adjusted terms and hence of *some* economic value. However, if we then isolate the CER estimates that are large, meaning in excess of the approximate 4 percent risk-free interest rate that prevailed during the 2020-2022 OOS period, we find that across all cryptocurrencies only 5 CER estimates exceed such a threshold. This means that converting the solid OOS performances shown by the R_{OS}^2 statistics into excess, abnormal performance at the level of each individual ticker remains challenging. Finally, while all the large CER exceeding 4 percent are associated with large and precisely estimated R_{OS}^2 coefficients (e.g., ADA, ETH, NEO, and MKR), there are many cases (e.g., BTG, DASH, DGB, GAS, LTC, and PAY) in which a strong statistical predictive performance fails to lead to very high CER for our OOS period.

As a summary, in Table 7, the average OOS R-squared are 6.60, 5.83, and 5.72 percent, respectively; under the assumption of independent samples, the first two estimates are imply values of 0.04 and 0.05 and are therefore statistically significant, while the latter estimate is characterized by a p-value of 0.06. Although the differences are small and never statistically significant, these results somewhat support the claim the two fundamental factors, especially the number of users, are more useful than sentiment-driven factors. Also in aggregate terms however, this strong predictability fails to map in extremely high economic value, as the average CER for the three prediction factors are 2.18, 2.22, and 2.18 percent per year, and hence rather homogeneous.

Similar patterns and qualitative results can be observed in Table 8 when the recursive OOS backtesting involves the Ex-BTC factors. For instance, with reference to the HashRate factor, the number of significant R_{OS}^2 coefficients simply declines from 39 to 38 when we go from the version of the factor in Table 7 to the one in Table 8, and the BTC data are excluded. The only exception concerns the Google Ex-BTC factor, which tends to underperform relative to the simple average in forecasting cryptocurrency returns. In fact, in the predictive regression based on sentiment, the estimated R_{OS}^2 is positive for 36 tickers out of 41 but it is significant only in 3 cases, which is even less than what one

Table 8: OOS predictive results for a cross-section of cryptocurrencies when the factor predictors are Ex-BTC

Ticker	User Factor Ex-BTC			Hashrate Factor Ex-BTC			Google Factor Ex-BTC		
	OOS R-square	p-value	CER ($\gamma = 3$)	OOS R-square	p-value	CER ($\gamma = 3$)	OOS R-square	p-value	CER ($\gamma = 3$)
ADA	11.30	0.00	6.08	10.65	0.00	5.68	-1.60	0.56	1.13
ANT	0.68	0.21	0.16	0.13	0.28	-0.13	0.66	0.93	0.07
BAT	3.96	0.05	2.46	3.02	0.06	1.98	0.45	0.51	0.10
BTC	6.61	0.03	2.05	2.76	0.02	2.35	2.34	0.01	1.98
BTG	10.82	0.00	0.67	10.29	0.00	0.43	0.25	0.24	0.05
CVC	1.37	0.10	0.18	0.07	0.19	0.02	0.24	0.40	0.21
DASH	3.58	0.01	0.06	7.87	0.01	-1.10	2.56	0.05	1.56
DCR	5.60	0.02	3.07	5.23	0.02	3.14	1.67	0.06	1.31
DGB	9.06	0.00	2.10	8.51	0.00	2.38	0.31	0.49	0.41
DOGE	2.46	0.04	2.09	1.68	0.07	2.56	0.90	0.51	1.46
DRGN	0.78	0.13	1.25	0.04	0.18	-1.63	0.86	0.08	0.08
ELF	4.77	0.01	1.80	4.10	0.01	1.82	1.38	0.83	0.07
ETC	6.42	0.01	1.68	5.04	0.02	0.52	0.99	0.41	2.00
ETH	9.28	0.00	6.10	6.90	0.00	6.48	1.38	0.94	0.07
FUN	0.28	0.38	0.25	1.27	0.41	0.38	1.72	0.97	0.42
GAS	13.22	0.02	0.02	15.33	0.03	0.32	-1.37	1.00	0.01
GNO	6.25	0.00	3.58	3.79	0.00	3.48	1.78	0.96	0.12
GNT	5.63	0.03	1.55	6.81	0.03	1.39	2.37	0.49	0.24
KNC	0.58	0.15	0.96	1.10	0.23	0.61	0.02	0.34	0.47
LEND	0.01	0.16	0.25	1.65	0.29	0.39	2.81	0.91	0.42
LINK	35.28	0.00	16.28	28.71	0.00	15.10	2.10	0.70	1.25
LTC	7.41	0.01	1.98	6.75	0.01	2.38	1.65	0.07	-0.56
MAID	1.64	0.42	-1.38	-1.98	0.37	-1.28	-1.54	0.98	0.21
MANA	2.82	0.05	1.63	1.72	0.08	1.31	0.83	0.86	0.26
MKR	14.07	0.00	14.07	11.60	0.00	13.87	9.98	0.00	12.37
NEO	12.74	0.00	4.94	11.36	0.00	4.22	-1.61	1.00	0.06
OMG	1.93	0.04	0.25	0.67	0.07	-0.06	1.90	0.09	0.08
PAY	1.73	0.08	0.03	2.68	0.04	0.02	1.08	0.08	0.16
POWR	3.89	0.04	0.17	3.21	0.04	0.00	1.21	0.81	0.14
PPT	0.55	0.37	0.08	-1.10	0.48	0.01	-1.49	0.93	0.00
REP	2.72	0.07	0.19	2.70	0.07	0.16	0.16	0.26	0.08
SNT	1.14	0.13	0.99	0.21	0.17	0.90	0.02	0.94	0.13
VTC	2.15	0.12	0.32	1.86	0.12	0.37	0.57	0.12	1.13
WAVES	4.00	0.01	0.73	4.42	0.01	0.78	2.13	0.68	0.04
WTC	5.83	0.03	2.00	4.41	0.01	-2.16	2.74	0.85	0.28
XEM	4.68	0.02	0.97	5.69	0.02	1.63	1.28	0.91	0.47
XLM	7.52	0.01	2.80	4.82	0.02	1.88	1.06	0.55	0.14
XRP	3.47	0.04	0.92	3.59	0.06	0.91	3.80	0.62	-7.65
XVG	5.60	0.02	0.94	3.85	0.04	0.50	0.20	0.12	1.24
ZEC	8.41	0.01	1.56	6.53	0.02	1.00	1.57	0.09	0.66
ZRX	4.07	0.02	2.08	2.32	0.04	1.51	2.30	0.92	0.66
Average	5.71	0.07	2.14	4.88	0.09	1.81	1.21	0.54	0.57

The table presents the OOS summary predictive performance results of the three factor-mimicking portfolio predictors isolated earlier when BTC returns are not used in their estimation, and applied to 41 cryptocurrencies. The table reports the cryptocurrency ticker, the OOS predictive percentage R-square, the associated p-value, and the annualized, percentage certainty equivalent for a mean-variance investor characterized by a risk aversion coefficient of 3. The OOS R-squared that are boldfaced are significant with a p-value of 5% or less.

would expect based in pure chance. The average R_{OS}^2 coefficients across cryptocurrencies are indeed 5.71 (p-value 0.07), 4.88 (p-value 0.09), and 1.21 (p-value 0.54) percent, and while the former two statistics are close to what found in Table 7, the latter is barely one quarter vs. the case in which predictions exploit all of BTC information content.

We close this Section on OOS predictability by exploring the economic value of a

Table 9: Results of backtesting the investment portfolio strategies

Portfolio Sorting Criterion	Weekly Return	Weekly SharpeRatio	Weekly % Alpha	p-value
User OOS R-square	-0.12	-0.03	-0.05	0.70
User OOS CER	0.38	0.09	0.23	0.05
Momentum OOS R-square	0.32	0.10	0.23	0.01
Momentum OOS CER	0.05	-0.16	-0.78	0.05
Google Sent. Factor OOS R-square	0.26	0.06	0.15	0.99
Google Sent. Factor OOS CER	0.62	0.21	0.53	0.00
User Factor OOS R-square	0.26	0.00	0.11	0.99
User Factor OOS CER	0.26	0.00	0.15	0.99
Hashrate Factor OOS R-square	0.06	0.11	0.14	0.70
Hashrate Factor OOS CER	0.90	0.23	0.47	0.00
Google Sent. Factor Ex-BTC OOS R-square	-0.19	-0.06	-0.10	0.03
Google Sent. Factor Ex-BTC OOS CER	-0.09	-0.09	0.00	0.05
User Factor Ex-BTC OOS R-square	0.05	0.06	0.18	0.68
User Factor Ex-BTC OOS CER	0.87	0.26	0.42	0.00
Hashrate Factor Ex-BTC OOS R-square	0.12	0.08	0.08	0.05
Hashrate Factor Ex-BTC OOS CER	0.80	0.24	0.42	0.00

The table reports the cumulative weekly returns, Sharpe ratios, estimated alphas, and the related p-value for portfolio strategies based on a long-short investment rule applied to predictive performance-based sorts of 41 cryptocurrencies. The alpha is calculated by regressing the realized returns of the various portfolio strategies on the returns on the CMBI benchmark. The analysis period covers the sample January 2020 - September 2022, for a total of 140 observations. Specifically, each strategy buys the 20 cryptocurrencies that can be better predicted and sells the 20 cryptocurrencies that are predicted worse within the sample. Significant statistics based on a 5% size test are boldfaced.

novel portfolio sorting strategy based on the results underlying Tables 7 and 8. Table 9 reports the average weekly return, the Sharpe ratio, and the realized portfolio alpha of a strategy that involve the systematic shorting of poorly predictable (e.g., characterized by a low or negative partial, recursive OOS squared) cryptocurrencies and buying those with better predictive accuracy, consistently with prior findings.²² The table shows that the long-short strategy exploiting the least effective factors (e.g., Google Ex-BTC, the User characteristic when sorting is based on R_{OS}^2 , and momentum when sorting is based on CER) yield negative performances. In general the strategies based on sorts that rely on the most successful factor mimicking predictors, especially those based on their CER lead to excellent total as well as risk-adjusted performances. For instance, ranking cryptocurrencies on the basis of their recursive OOS CER yields a long-short portfolio which gives as much as 0.90 percent per week, with a weekly Sharpe ratio of 0.23 and

²²The alpha of the strategy is estimated in a regression in which systematic risk is captured by the returns of the the CMBI portfolio, also to avoid redundancy with respect to the other data used in factor construction and earlier tests.

highly significant alpha of 0.47 percent. More generally, Table 9 shows that in the case of nine strategies out of 16, sorting on the basis of realized OOS CER leads to stronger performance vs. a R_{OS}^2 -led sorting, highlighting the importance of considering risk preferences when constructing cryptocurrency portfolios. Finally, the large Sharpe ratios and average abnormal returns obtained from this long-short strategy based on fundamental blockchain-based predictability suggest the possibility that technological features such as the power and network spread of blockchains may have become pervasive enough to affect the very nature of the fundamental pricing measure characterizing the financial markets.

5 Conclusion

The objective of this paper is to investigate the predictability of cryptocurrency returns, structuring the analysis into two distinct steps. Firstly, we examine the contemporaneous relationships between fundamental factors and cryptocurrency returns, what we refer to as the in-sample analysis. Secondly, we introduce a novel approach based on the investigation of the behavior of fundamental factors as predictors of cryptocurrency returns. Additionally, we develop and assess the realized recursive performance of portfolios that invest in cryptocurrencies with higher predictability while shorting those with lower predictability. The strong results obtained by such long-short strategies emphasize the economic value of the predictive power of blockchain fundamentals and sentiment indicators in the cryptocurrency markets.

To perform the in-sample analysis, we construct three factor mimicking portfolios, using the returns of the cryptocurrencies themselves, based on the fundamental characteristics of the blockchains: active users (Users), hashing power (HashRate), and Google Trends index score. The analysis demonstrates the effectiveness of these factors in explaining cryptocurrency returns, consistently with earlier findings in the literature ([Bhambhani et al., 2023](#)). The empirical findings reveal that even they are taken in isolation, the fundamental factors outperformed a naive historical average in terms of explanatory power for cryptocurrency returns.

In the out-of-sample analysis, the factor mimicking portfolio predictors yield significant cumulative forecast improvements for cryptocurrency returns, with overperformance spreads ranging between 10 and 35 percent by the end of the OOS period. Across 41 cryptocurrencies, the OOS values are predominantly positive, with fundamental predic-

tors (Users and HashRate) performing slightly better than sentiment predictors, although their economic value, measured by CER, remains modest. Despite the existence of strong statistical predictability across models, when they are considered individually, only a few cryptocurrencies (e.g., ADA, ETH) convert this into economically meaningful excess returns above the risk-free rate. This highlights the challenges in translating predictive power into realized abnormal performance. We also find that the estimated alphas obtained from the long-short portfolios constructed on the basis of cryptocurrency sorting reflecting the predictive performances, records a significant increase when the individual cryptocurrencies are ranked according to their certainty equivalent, as opposed to those obtained by sorting them by R^2 . This further stresses the importance of incorporating risk-adjusted metrics, such as the CER, in the assessment of predictive performance, thereby offering valuable insights for the optimization of investment strategies. These results support the view of cryptocurrencies as digital commodities that require substantial computational power and electricity for their creation, akin to the extraction of physical commodities. like gold. Moreover, the production of cryptocurrencies would not generate new value-added but simply redistribute existing value within the global economy.²³

Our paper leaves open several avenues of future exploration. First, we believe that constructing fundamental factors based on alternative blockchain characteristics, such as network value to transaction (NTV, the ratio between a cryptocurrency's market capitalization and its transaction activity, the total value of all on-chain transactions, see [Sakkas and Urquhart \(2024\)](#)) may warrant consideration. This is especially relevant given the market shift towards blockchains that deviate from the PoF Work consensus mechanism, which is a necessary condition to perform hashrate calculation, in favor of more energy-efficient consensus algorithms, such as PoS. The results of our study also suggest that investors and researchers should consider directly (e.g., within a recursive sorting framework applied to different cryptocurrencies, as we do) incorporating risk-adjusted metrics, such as the CER, in addition to traditional predictive measures, like the OOS R^2 , when assessing the potential of cryptocurrency investments. These considerations could provide a more robust framework for forecasting returns, especially as the cryptocurrency market evolves and becomes increasingly complex. Finally, we believe that there is considerable

²³Recently, the Commodity Futures Trading Commission (CFTC) has asserted its jurisdiction over the derivatives written on Bitcoin by classifying it as a commodity, thereby overseeing its derivatives and related financial products. This regulatory stance underscores the importance of understanding Bitcoin's role within the commodity framework to ensure appropriate oversight and consumer protection.

potential in providing an extension of the analysis to a broader range of cryptocurrencies, with the goal of developing more general fundamental factors, although such an extension may only become feasible in the coming years as the market continues to mature.

References

- Allen, F., Gu, X., and Jagtiani, J. (2022). Fintech, cryptocurrencies, and cbdc: Financial structural transformation in china. *Journal of International Money and Finance*, 124:102625.
- Baker, M. and Wurgler, J. (2006). Investor sentiment and the cross-section of stock returns. *Journal of Finance*, 61(4):1645–1680.
- Bali, T. G., Engle, R. F., and Murray, S. (2016). *Empirical asset pricing: The cross section of stock returns*. John Wiley & Sons.
- Barillas, F. and Shanken, J. (2017). Which alpha? *Review of Financial Studies*, 30(4):1316–1338.
- Baur, D. G., Hong, K., and Lee, A. D. (2018). Bitcoin: Medium of exchange or speculative assets? *Journal of International Financial Markets, Institutions and Money*, 54:177–189.
- Bhambhwani, S. M., Delikouras, S., and Korniotis, G. M. (2023). Blockchain characteristics and cryptocurrency returns. *Journal of International Financial Markets, Institutions and Money*, 86:101788.
- Biais, B., Bisiere, C., Bouvard, M., Casamatta, C., and Menkveld, A. J. (2023). Equilibrium bitcoin pricing. *Journal of Finance*, 78(2):967–1014.
- Bianchi, D., Babiak, M., and Dickerson, A. (2022). Trading volume and liquidity provision in cryptocurrency markets. *Journal of Banking & Finance*, 142:106547.
- Bianchi, D., Guidolin, M., and Pedio, M. (2023). The dynamics of returns predictability in cryptocurrency markets. *European Journal of Finance*, 29(6):583–611.
- Campbell, J. Y. and Shiller, R. J. (1988). Stock prices, earnings, and expected dividends. *Journal of Finance*, 43(3):661–676.
- Campbell, J. Y. and Thompson, S. B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531.
- Cattaneo, M. D., Crump, R. K., Farrell, M. H., and Schaumburg, E. (2020). Characteristic-sorted portfolios: Estimation and inference. *Review of Economics and Statistics*, 102(3):531–551.

- Chaudhry, N. and Yousaf, M. M. (2018). Consensus algorithms in blockchain: Comparative analysis, challenges and opportunities. In *2018 12th international conference on open source systems and technologies (ICOSST)*, pages 54–63. IEEE.
- Chiu, J. and Koepl, T. V. (2022). The economics of cryptocurrency: Bitcoin and beyond. *Canadian Journal of Economics*, 55(4):1762–1798.
- Cochrane, J. (2005). Financial markets and the real economy. *Foundations and Trends® in Finance*, 1(1):1–101.
- Cong, L. W., Li, Y., and Wang, N. (2021). Tokenomics: Dynamic adoption and valuation. *Review of Financial Studies*, 34(3):1105–1155.
- Diebold, F. X. and Mariano, R. S. (2002). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 20(1):134–144.
- Dziembowski, S., Faust, S., Kolmogorov, V., and Pietrzak, K. (2015). Proofs of space. In *Annual Cryptology Conference*, pages 585–605. Springer.
- Easley, D., O’Hara, M., and Basu, S. (2019). From mining to markets: The evolution of bitcoin transaction fees. *Journal of Financial Economics*, 134(1):91–109.
- Foley, S., Karlsen, J. R., and Putniņš, T. J. (2019). Sex, drugs, and bitcoin: How much illegal activity is financed through cryptocurrencies? *Review of Financial Studies*, 32(5):1798–1853.
- Ginsberg, J., Mohebbi, M. H., Patel, R. S., Brammer, L., Smolinski, M. S., and Brilliant, L. (2009). Detecting influenza epidemics using search engine query data. *Nature*, 457(7232):1012–1014.
- Gordon, M. J. (1959). Dividends, earnings, and stock prices. *Review of Economics and Statistics*, pages 99–105.
- Guidolin, M., Orlov, A. G., and Pedio, M. (2018). How good can heuristic-based forecasts be? a comparative performance of econometric and heuristic models for uk and us asset returns. *Quantitative Finance*, 18(1):139–169.
- Härdle, W. K., Harvey, C. R., and Reule, R. C. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2):181–208.
- Jo, H., Park, H., and Shefrin, H. (2020). Bitcoin and sentiment. *Journal of Futures Markets*, 40(12):1861–1879.
- Kagkadis, A., Lohre, H., Nolte, I., and Vasilas, N. (2023). Power sorting. *Working paper, Lancaster University*.
- Kan, R. and Zhou, G. (2007). Optimal portfolio choice with parameter uncertainty. *Journal of Financial and Quantitative Analysis*, 42(3):621–656.

- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Li, Y., Zhang, W., Urquhart, A., and Wang, P. (2022). The role of media coverage in the bubble formation: evidence from the bitcoin market. *Journal of International Financial Markets, Institutions and Money*, 80:101629.
- Liu, Y. and Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6):2689–2727.
- Liu, Y., Tsyvinski, A., and Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2):1133–1177.
- Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319.
- Marquering, W. and Verbeek, M. (2004). The economic value of predicting stock index returns and volatility. *Journal of Financial and Quantitative Analysis*, 39(2):407–429.
- Mohamad, A. and Stavroyiannis, S. (2022). Do birds of a feather flock together? evidence from time-varying herding behaviour of bitcoin and foreign exchange majors during covid-19. *Journal of International Financial Markets, Institutions and Money*, 80:101646.
- Mokni, K., Bouteska, A., and Nakhli, M. S. (2022). Investor sentiment and bitcoin relationship: A quantile-based analysis. *The North American Journal of Economics and Finance*, 60:101657.
- Narayanan, A. (2016). *Bitcoin and cryptocurrency technologies: a comprehensive introduction*. Princeton University Press.
- Noda, S., Okumura, K., and Hashimoto, Y. (2020). An economic analysis of difficulty adjustment algorithms in proof-of-work blockchain systems. *EC*, 20:13–17.
- Pagnotta, E. and Buraschi, A. (2018). An equilibrium valuation of bitcoin and decentralized network assets. *Working paper*.
- Pagnotta, E. S. (2022). Decentralizing money: Bitcoin prices and blockchain security. *Review of Financial Studies*, 35(2):866–907.
- Prat, J. and Walter, B. (2021). An equilibrium model of the market for bitcoin mining. *Journal of Political Economy*, 129(8):2415–2452.
- Rapach, D. E., Strauss, J. K., and Zhou, G. (2010). Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *Review of Financial Studies*, 23(2):821–862.

- Sakariyahu, R., Lawal, R., Adigun, R., Paterson, A., and Johan, S. (2024). One crash, too many: Global uncertainty, sentiment factors and cryptocurrency market. *Journal of International Financial Markets, Institutions and Money*, 94:102028.
- Sakkas, A. and Urquhart, A. (2024). Blockchain factors. *Journal of International Financial Markets, Institutions and Money*, 94:102012.
- Selgin, G. (2015). Synthetic commodity money. *Journal of Financial Stability*, 17:92–99.
- Shiller, R. J. (1981). Do stock prices move too much to be justified by subsequent changes in dividends? *American Economic Review*, 71(3):421–436.
- Sockin, M. and Xiong, W. (2023). Decentralization through tokenization. *Journal of Finance*, 78(1):247–299.
- Stambaugh, R. F., Yu, J., and Yuan, Y. (2012). The short of it: Investor sentiment and anomalies. *Journal of Financial Economics*, 104(2):288–302.
- Telford, T. (2019). After founder’s sudden death, cryptocurrency exchange can’t access \$190 million in holdings. *The Washington Post*. Published February 4, 2019.
- Valdeolmillos, D., Mezquita, Y., González-Briones, A., Prieto, J., and Corchado, J. M. (2020). Blockchain technology: a review of the current challenges of cryptocurrency. In *Blockchain and Applications: International Congress*, pages 153–160. Springer.
- Wachter, J. A. and Warusawitharana, M. (2009). Predictable returns and asset allocation: Should a skeptical investor time the market? *Journal of Econometrics*, 148(2):162–178.
- Welch, I. and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508.
- West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica*, 64(5):1067–1084.
- Xu, L. and Kinkyo, T. (2023). Hedging effectiveness of bitcoin and gold: Evidence from g7 stock markets. *Journal of International Financial Markets, Institutions and Money*, 85:101764.
- Yermack, D. (2024). Is bitcoin a real currency? an economic appraisal. In *Handbook of Digital Currency*, pages 29–40. Elsevier.
- Yousaf, I., Abrar, A., and Yarovaya, L. (2023). Decentralized and centralized exchanges: Which digital tokens pose a greater contagion risk? *Journal of International Financial Markets, Institutions and Money*, 89:101881.
- Zheng, Z., Xie, S., Dai, H.-N., Chen, X., and Wang, H. (2018). Blockchain challenges and opportunities: A survey. *International Journal of Web and Grid Services*, 14(4):352–375.